

Technical Specification: Interaction Overview Diagram

Module: System Integration & Administrative Control

Prepared by: Eneida Balliu & Ajla Alcani

Requirement Reference: UC_40 (Integration Layer)

Document Objective

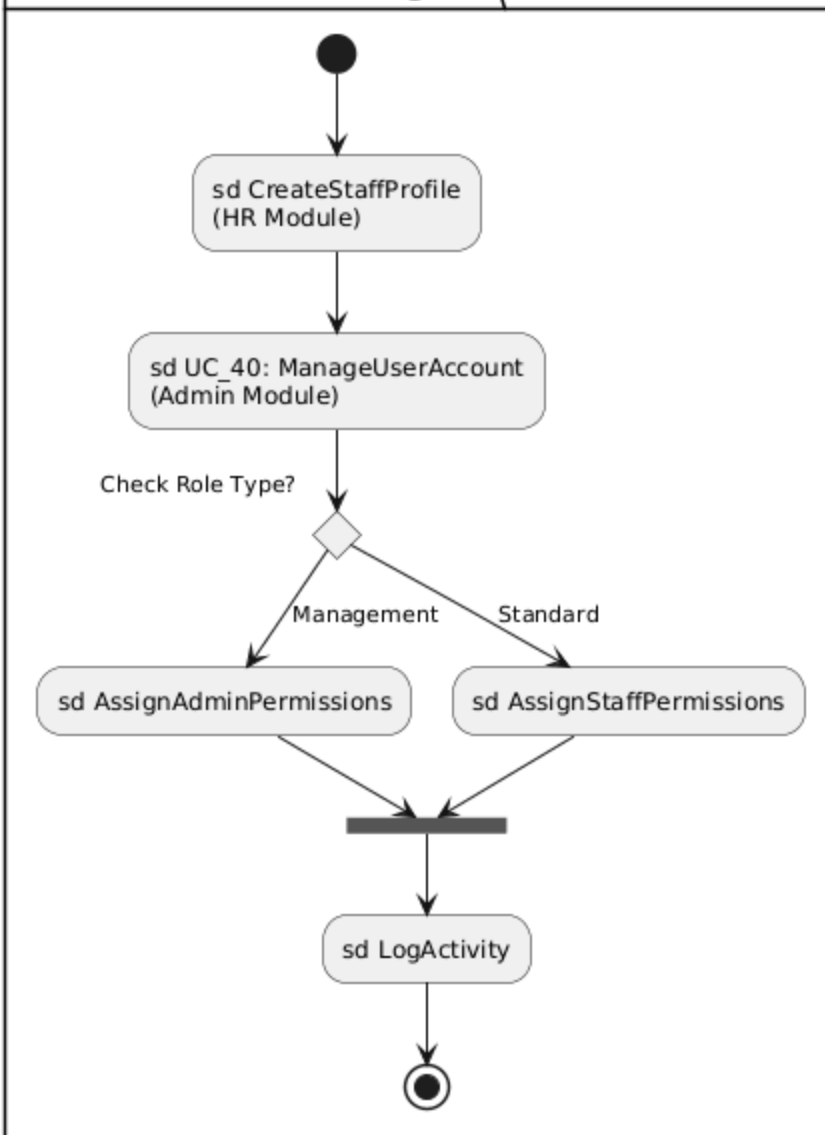
The purpose of this diagram is to provide a macro-perspective of the system's behavior by coordinating multiple interaction models into a single unified workflow. While Collaboration Diagrams show detailed messaging, the Interaction Overview Diagram focuses on the "Big Picture" logic—specifically the hand-off between the Human Resources (HR) module and the System Administration module.

Technical Scope

This model encapsulates the following system transitions:

- *Inter-Module Hand-off: The transition from HR profile creation to Digital Account management.*
- *Interaction Occurrences: The use of sd (Sequence/Interaction) frames to reference externalized logic without cluttering the main flow.*
- *Conditional Synchronization: The routing of security permissions based on user role guards.*

Interaction Overview Diagram



Technical Specification: Interaction Overview Model (Staff Onboarding)

Objective: This diagram illustrates the high-level orchestration of the Staff Onboarding Process, specifically mapping the integration between the Human Resources (HR) and System Administration modules.

Core Components & Logic:

- Interaction References (sd Blocks): Each rectangular box (e.g., sd UC_40: ManageUserAccount) represents an encapsulated Interaction Use. This indicates that the diagram references complex message flows (Sequence or Collaboration diagrams) previously defined, maintaining modularity and a clean architectural overview.
- Decision and Merge Nodes: The model utilizes a Decision Diamond to branch the workflow based on user roles (Guard Conditions: [Management] vs. [Standard]).
- Logical Synchronization: A Merge Node is implemented following the permission-assignment phase. This ensures that regardless of the chosen path, the control flow is synchronized to perform the final LogActivity interaction.
- Final State Reachability: By closing the "Management" branch and merging it back into the main flow, the diagram satisfies the UML requirement for Reachability, ensuring every possible scenario concludes at the Final Node.

Architectural Value: This model provides traceability across system boundaries. It proves that the administrative account management (UC_40) is a prerequisite for permission assignment and that all actions are subject to a unified system logging procedure for security compliance.

Technical Specification: Interaction Overview Diagram

Module: System Integration & Administrative Control

Prepared by: Albina Blloshmi and Betül Kaan

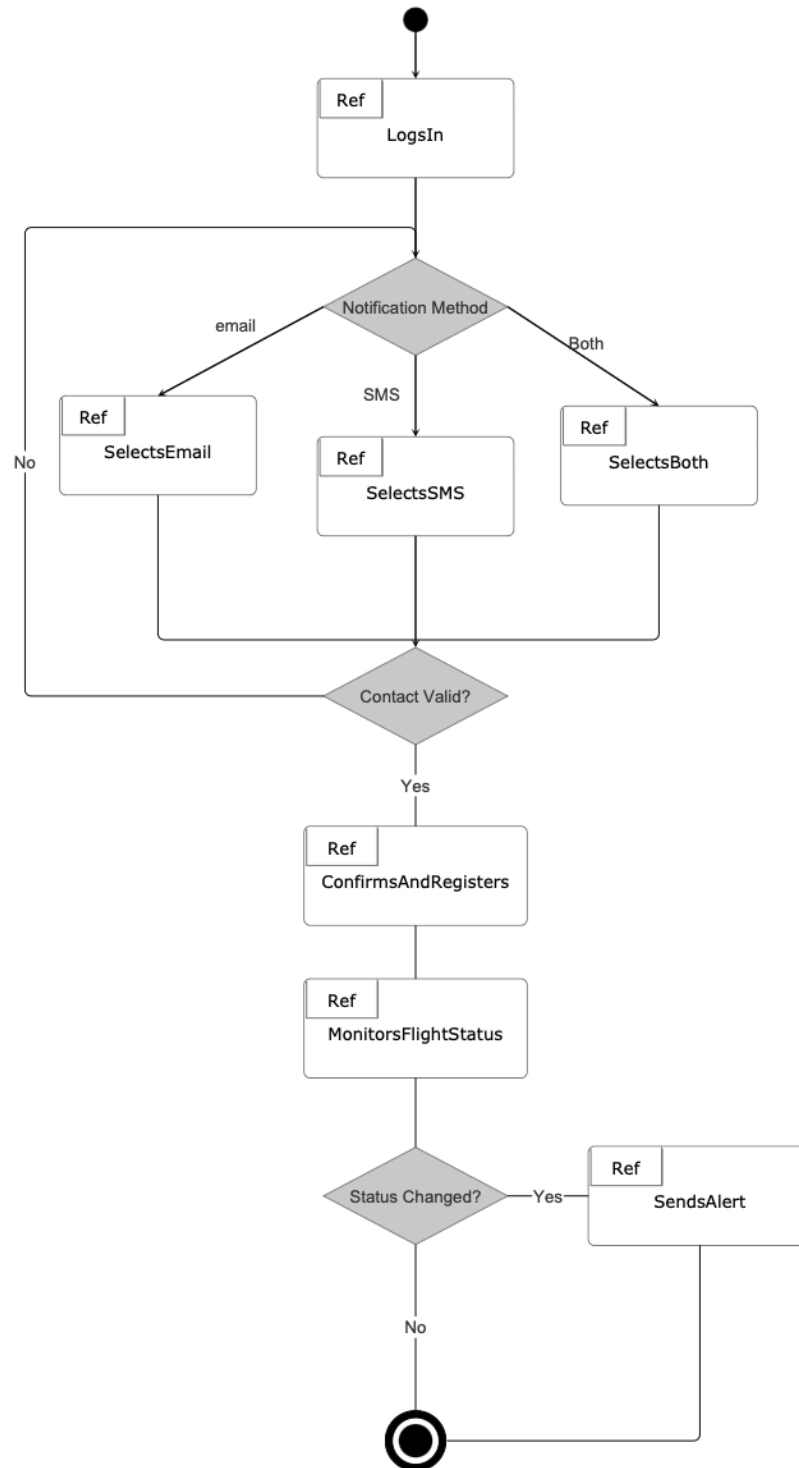
Requirement Reference: UC_07 (Flight Notification)

Document Objective

The purpose of this diagram is to provide a macro-perspective of the system's behavior by coordinating multiple interaction models into a single unified workflow. While Collaboration Diagrams show detailed messaging, the Interaction Overview Diagram focuses on the "Big Picture" logic—specifically the hand-off between the Human Resources (HR) module and the System Administration module.

Technical Scope

- **Passenger-to-System Hand-off:** The transition from passenger-initiated actions (authentication, flight selection, and contact method preference) to system-owned processes (contact validation, alert registration, and real-time flight monitoring).
- **Interaction Occurrences:** The use of ref frames to reference externalized notification logic — such as SelectsEmail, SelectsSMS, and SelectsBoth — without cluttering the main orchestration flow, preserving modularity across subordinate sequence diagrams.
- **Conditional Branching on Notification Method:** The routing of the subscription flow across three mutually exclusive paths based on the passenger's chosen delivery channel, governed by guard conditions [EMAIL], [SMS], and [BOTH].
- **Contact Validation Gate:** The enforcement of a loopback mechanism via the ContactValid? decision node, ensuring that malformed contact details are intercepted before registration is committed to the system.
- **Monitoring and Alert Dispatch:** The transition from a one-time registration event to a continuous monitoring state, with conditional alert delivery governed by the StatusChanged? guard, ensuring notifications are only dispatched upon a verified flight status update.



Process Explanation

1. Start

* The process begins when the user enters the system.

2. Login
 - * The user logs into the application or platform.
3. Choose Notification Method
 - * The system asks the user to select how they want to receive notifications.
 - * The user can choose:
 - * Email
 - * SMS
 - * Both Email and SMS
4. Notification Selection
 - * Depending on the selected option:
 - * SelectsEmail → notifications will be sent through email.
 - * SelectsSMS → notifications will be sent through SMS.
 - * SelectsBoth → notifications will be sent through both channels.
5. Contact Validation
 - * The system checks whether the provided contact information is valid.
 - * Examples:
 - * valid email format
 - * active phone number
 - * If the contact information is invalid, the process loops back so the user can choose or re-enter the notification method/details again.
 - * If the contact information is valid, the process continues.
6. Confirms and Registers
 - * The user confirms their information.
 - * The system registers the user successfully for flight notifications.
7. Monitor Flight Status
 - * The system continuously monitors the status of the selected flight.
8. Check if Status Changed
 - * The system checks for any changes in the flight status, such as:
 - * delays
 - * cancellations
 - * gate changes
 - * departure time updates
9. Send Alert
 - * If the flight status changes, the system sends an alert to the user using the selected notification method(s).
10. End Process
 - * If there are no status changes, or after alerts are sent, the workflow reaches the end state.

Overall Purpose

The diagram represents a notification management system that allows users to receive real-time flight updates through email, SMS, or both communication methods.

Technical Specification: Interaction Overview Diagram

Module: Flight Operations & Security

Prepared by: Alvi Elezi & Ani Velo

Requirement Reference: UC_SEC_01-03; UC_01, 02, 05, 06

Document Objective

This diagram visualizes the high-level control flow between the Security and Flight Operations modules, focusing on the orchestration of multiple use cases. It establishes the logical dependencies and safety protocols required to clear a flight for dispatch, bridging the gap between individual sequence models and the complete operational lifecycle.

Technical Scope

The specification covers the coordination between the Security and Flight Operations Controllers, specifically addressing authentication, clearance verification, and iterative delay management. It details concurrent processes like parallel passenger screening and boarding, while ensuring that all security incidents and final dispatch actions are recorded within the entity layer for compliance.

Interaction overview diagram:

